

MicroPythonOS

FOSDEM – Sunday 1-2-2026

whoami



- Thomas Farstrike
- Creator of MPOS
- Background:
 - Kernel development
 - Embedded Linux
 - C/C++

Origins



- One Year Ago
- Building new device
- Need for modern ESP32 Touch OS
- ...with AppStore
- Didn't exist!

Create it in C/C++ ?



- Known territory
- Most ESP32 libraries
- Lightweight
- Logical choice
- Why not?

Create it in C/C++ ?



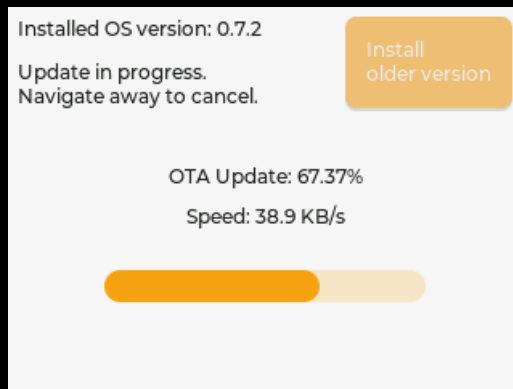
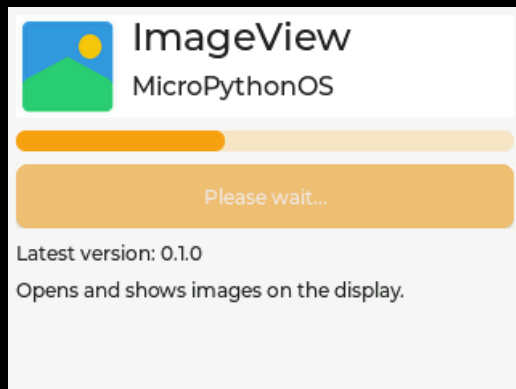
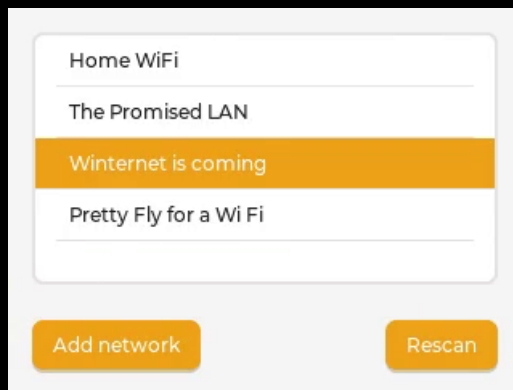
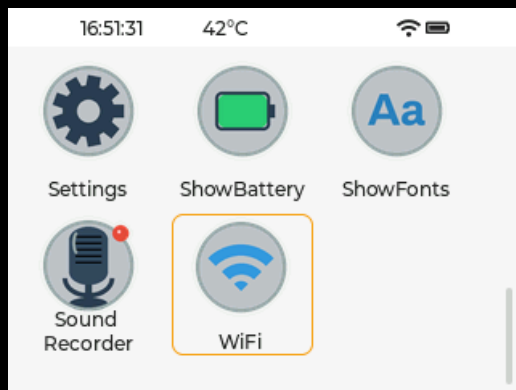
- Limited MMU
- Custom compilation
- Unforgiving
- Difficult to create OS
- Not easy for app developers

Create it in MicroPython!



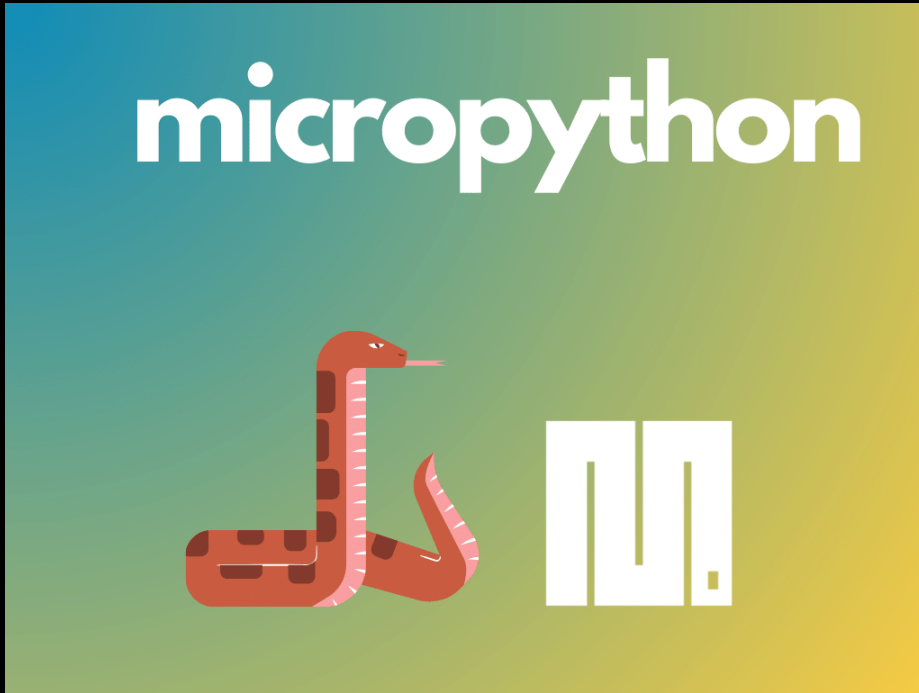
- Dynamic loading
- Scripts (.py)
- Binaries (.mpy)
- Modern language
- Easy for app developers

Proving the concept



- Launcher
- WiFi + Auto-Connect
- AppStore
- OSUpdate
- Settings
- About

MicroPython



- Born 2014, ESP 2017
- Surprisingly fast
- Many libraries
- Also desktop (native)
- Quick Coding Cycle

LVGL makes it beautiful



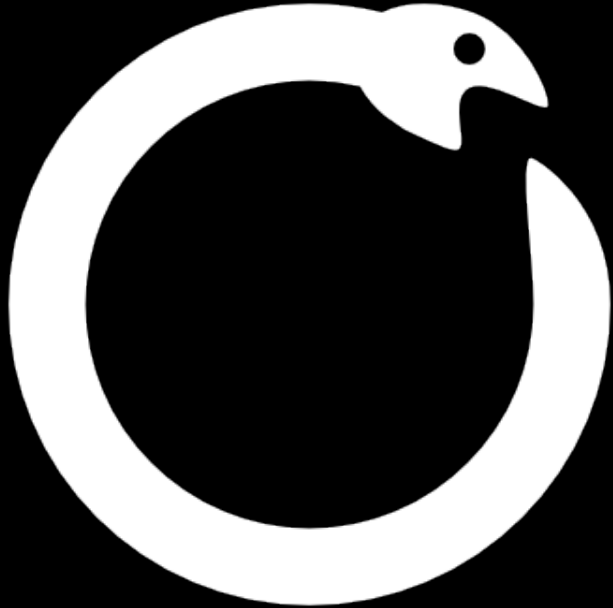
- Light and Versatile
- 30+ Widgets
- Animations
- Image Decoders
- UI Editor
- MicroPython bindings:
Credit: lvgl_micropython

Lessons from Android



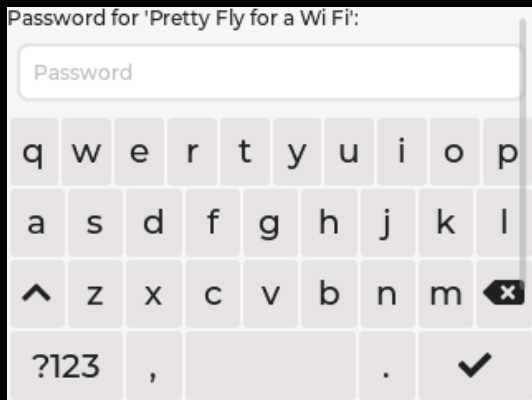
- Create once, run on all (compatible) devices
- Easy app development
- Frameworks
- User Interface

Why MicroPythonOS



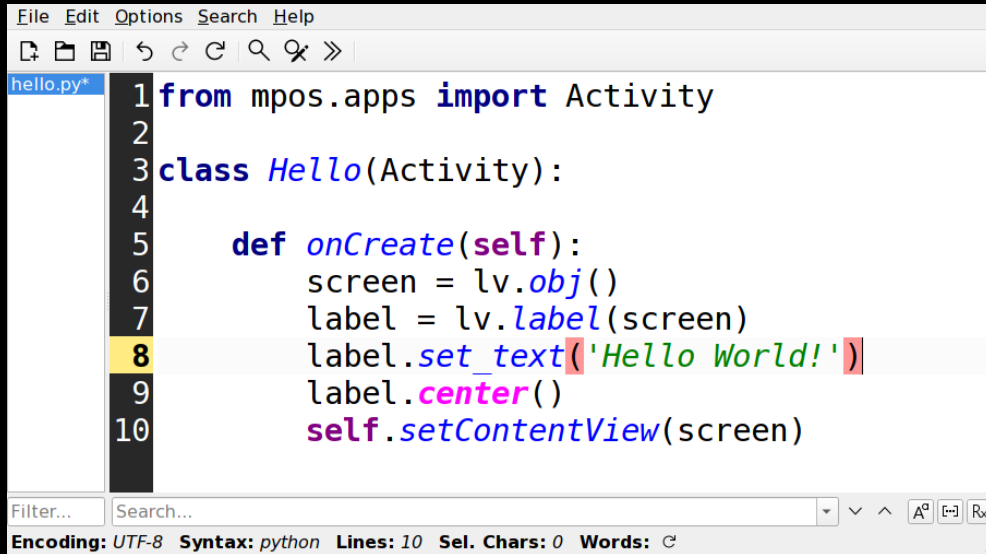
- Very Lightweight
- Python
- Boots in 3 seconds
- Public development
- Frequent releases
- Runs native on desktop
- Up-and-coming

Innovations



- MposKeyboard +50% buttons
- Swipe gestures for navigation
- Optimize QR decoding (960x960)
- v4l2 camera driver for MP
- Joystick for focus direction
- python-nostr ported to micropython-nostr
- Extend secp256k1 MP binding for ECDH

Familiar app life cycle



```
File Edit Options Search Help
[Icons]
hello.py*
1 from mpos.apps import Activity
2
3 class Hello(Activity):
4
5     def onCreate(self):
6         screen = lv.obj()
7         label = lv.label(screen)
8         label.set_text('Hello World!')
9         label.center()
10        self.setContentView(screen)
Filter... Search...
Encoding: UTF-8 Syntax: python Lines: 10 Sel. Chars: 0 Words: 6
```

- Create Screen
- Add Label
- Set Label Text
- Center Label
- Activate Screen

Familiar MANIFEST.JSON

```
1 {  
2 "name": "HelloWorld",  
3 "publisher": "MicroPythonOS",  
4 "short_description": "Minimal app",  
5 "long_description": "Demonstrates the simplest app.",  
6 "full_name": "com.micropythonos.helloworld",  
7 "version": "0.0.2",  
8 "category": "development",  
9 "activities": [  
10     {  
11         "entrypoint": "assets/hello.py",  
12         "classname": "Hello",  
13         "intent_filters": [  
14             {  
15                 "action": "main",  
16                 "category": "launcher"  
17             }  
18         ]  
19     }  
20 ]  
21 }
```

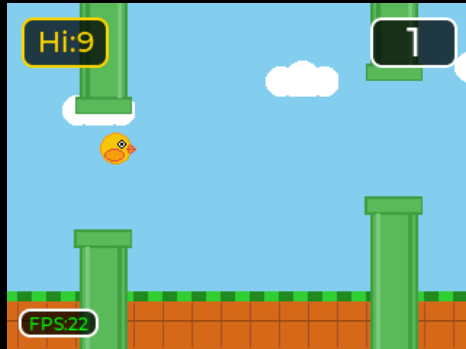
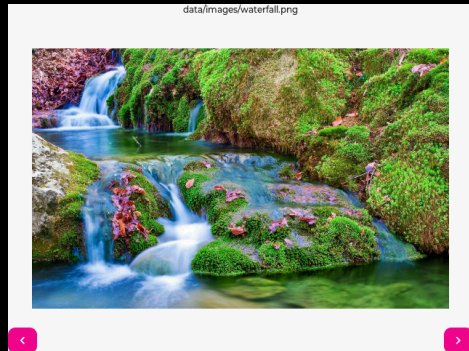
- Name
- Publisher
- Descriptions
- Version
- Entrypoints

Familiar file structure

```
apps/com.micropythonos.helloworld/  
├── assets  
│   └── hello.py  
├── META-INF  
│   └── MANIFEST.JSON  
└── res  
    ├── mipmap-mdpi  
    │   └── icon_64x64.png
```

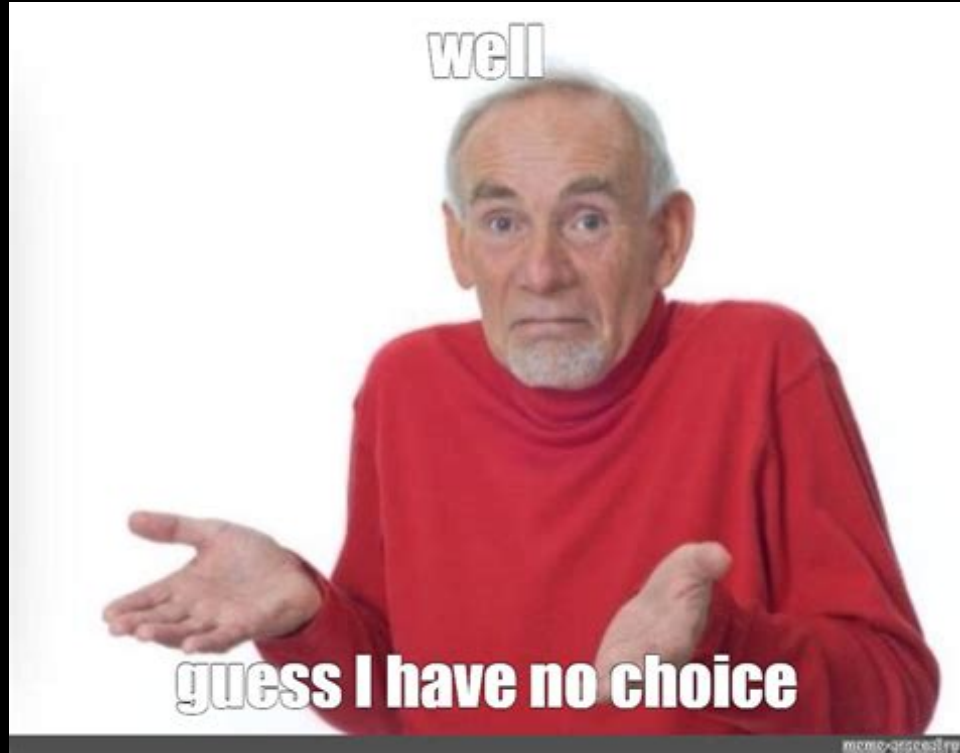
- Code (hello.py)
- MANIFEST.JSON
- Icon(s)
- Zipped to .mpk

Apps in the AppStore



- LightningPiggy
- Camera (with QR)
- Image Viewer
- Games
- Music Player
- Sound Recorder
- + 8 more

Why build a MicroPythonOS app



- Fun
- Educational
- Easy (or Hard)
- Supportive community
- $1/1 > 1/100$
- Growing niche

Good documentation

MicroPythonOS Documentation

MicroPythonOS Documentation

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BatteryManager

BuildInfo

ConnectivityManager

DeviceInfo

DisplayMetrics

DownloadManager

InputManager

LightsManager

Preferences

SensorManager

SettingActivity

SettingsActivity

TaskManager

TimeZone

WidgetAnimator

WifiService

Architecture

MicroPythonOS Documentation

Welcome to the official documentation for **MicroPythonOS**, a lightweight and versatile operating system built entirely in MicroPython. Designed for microcontrollers like the ESP32 and desktop systems, MicroPythonOS offers a modern, Android-inspired interface with an app ecosystem, App Store, and Over-The-Air (OTA) updates.

To see MicroPythonOS in action, check out the screenshots and screen videos on the [main website](#).

About MicroPythonOS

MicroPythonOS is tailored for developers and innovators building IoT devices, smart wearables, educational tools, and more. Its key features include:

- **Native MicroPython:** Write apps and drivers in MicroPython for rapid development.
- **Cross-Platform:** Runs on ESP32 microcontrollers, Linux desktops, and potentially MacOS and Raspberry Pi.
- **Modern UI:** Android-like touch screen interface with gestures.
- **App Ecosystem:** Built-in apps and an App Store for easy expansion.
- **OTA Updates:** Seamless system and app updates.

This documentation provides everything you need to get started, understand the architecture, build the system, and develop apps.

Who This Is For

- **Developers:** Learn how to install, build, and extend MicroPythonOS with new apps and features.
- **Managers:** Explore the system's capabilities, supported hardware, and ready-to-use features for prototyping and production.

Getting Started

New to MicroPythonOS? Start [here](#).

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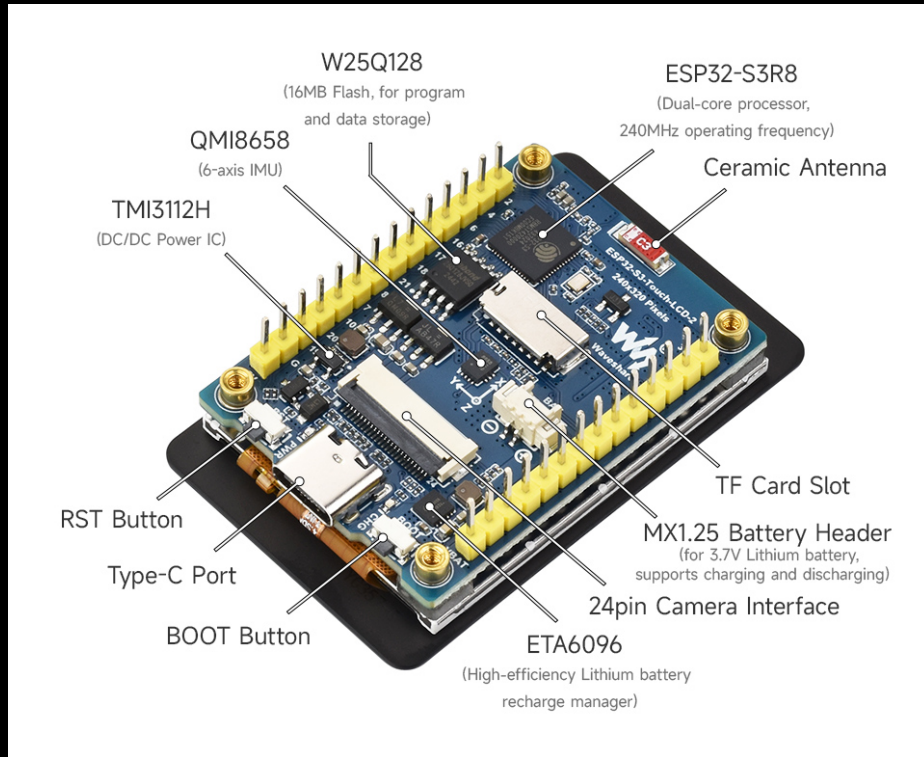
- All frameworks documented
- Updated regularly

<https://docs.MicroPythonOS.com>

Supported hardware

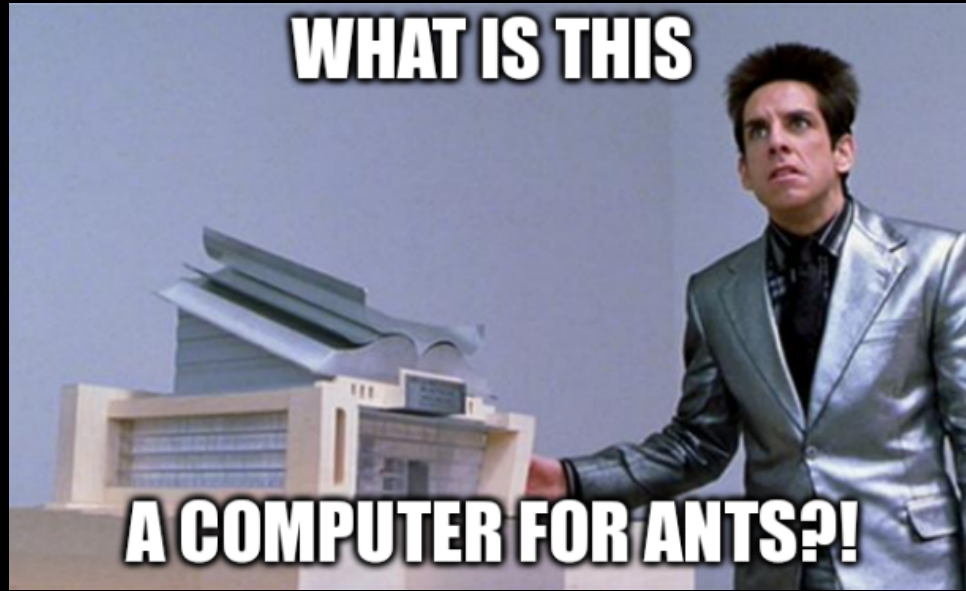
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First board: Waveshare S3 Touch 2



- 320x240
- 2 inch screen
- Capacitive Touch
- Camera 5MP
- Accelerometer (IMU)
- MicroSD Card
- Battery support

Hardware Specs



- 300KB SDRAM
- 8MB PSRAM
- 16MB ROM
- 320x240 (QVGA)

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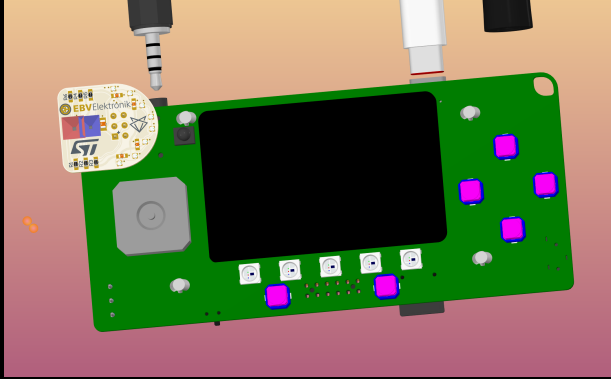
=~ IBM PS/2 (1981)

Second board: Fri3d Camp 2024



- 296x240
- 2 inch screen
- IMU
- Buzzer
- Multicolor LEDs
- Expansions

Third board: Fri3d Camp 2026



- 320x240 2 inch display
- Capacitive Touch!
- Audio Jack
- MicroPythonOS
- Retro-Go
 - Doom
 - Gameboy
 - NES

Next boards



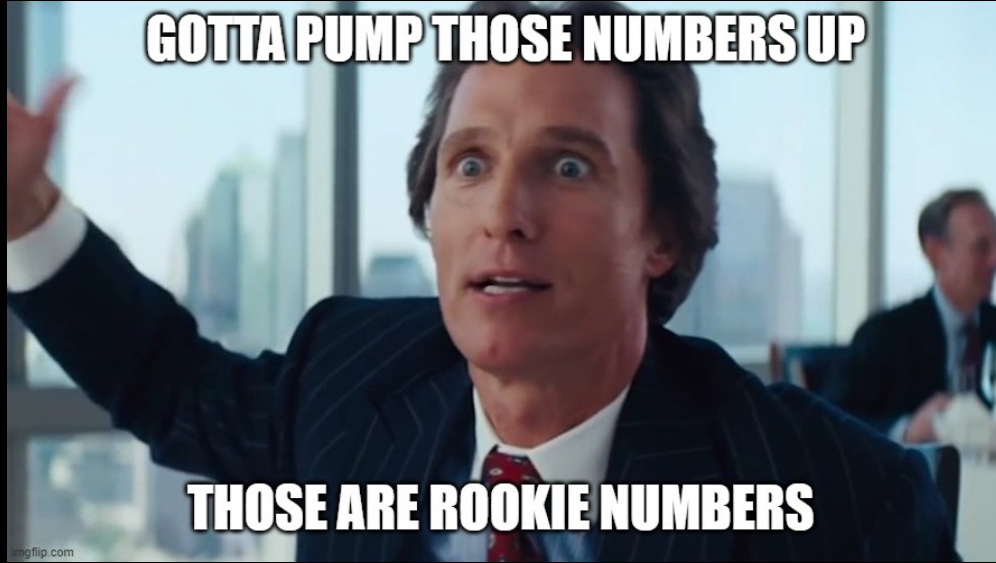
- Manufacturers are interested – adds value
- Samples received
- Different form factors
- Porting for free
 - Hurry if you want it

Live Demo



- You are looking at it.
- Yes, really!
- You were in it...
- the whole time!

MicroPythonOS in Numbers



- Core Contributors: 4
- Watching: 6
- Apps: 15
- Forks: 20
- Community Chat: 53
- GitHub stars: 319
- Code commits: 2000

Roadmap



- Factory installs
- More frameworks
- API 1.0
- More drivers
- More boards
- Community driven

Next Steps: free stuff



- Run it on desktop
- Prebuilt binaries:
 - Linux
 - MacOS
- Build an app

Next Steps: buy stuff



- Get a supported device
- Flash it
- Install your app
- Publish in AppStore
 - Multiple backends

Easy Installation

Install MicroPythonOS

Click the button below to flash [MicroPythonOS](#) firmware onto your ESP device using WebSerial. Ensure your device is connected via USB and you're using a supported browser (e.g., Chrome or Edge).

Don't have a device yet? Buy a supported one, like the [Waveshare ESP32-S3-Touch-LCD-2](#). Or check out [the docs](#) to see how you can run it on your desktop PC.

Select Version

- Unified for all ESP32-S3 devices: Waveshare and Fri3d Camp 2024
Badge: version 0.7.1

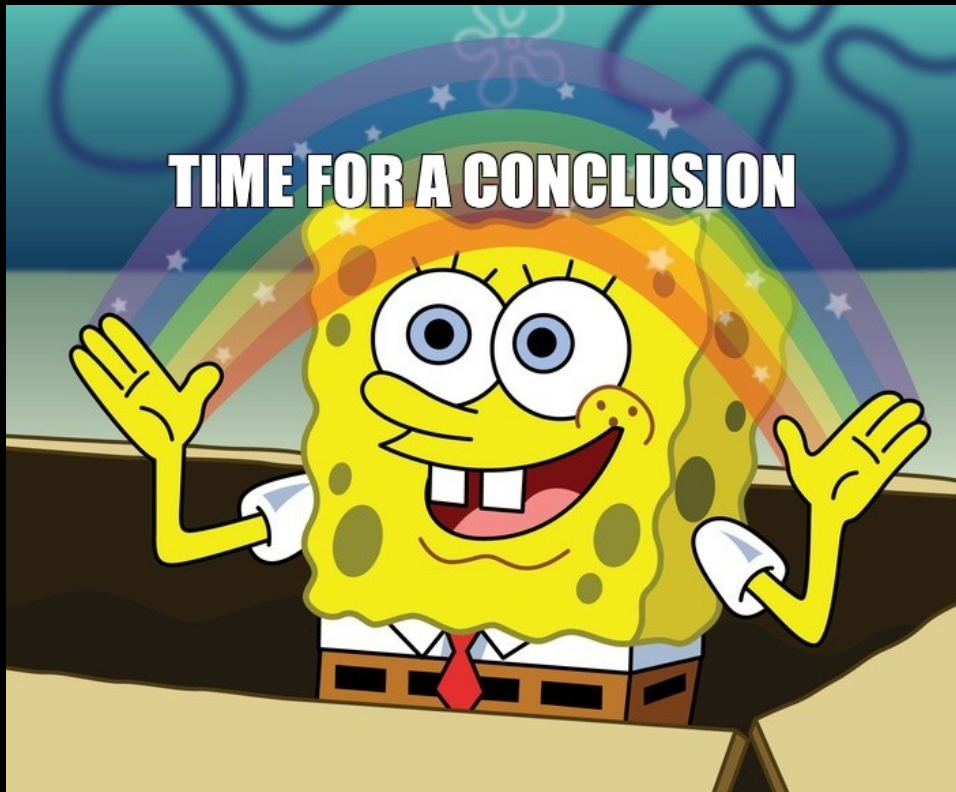
Show old releases ▼

Connect and Install

- WebSerial installer
- One firmware to rule them all

<https://install.MicroPythonOS.com>

Conclusion



- MicroPythonOS is waiting for you
- Join the chat
<https://chat.MicroPythonOS.com>
- Clone the repo
- Questions?

MicroPythonOS

Thanks FOSDEM and you!